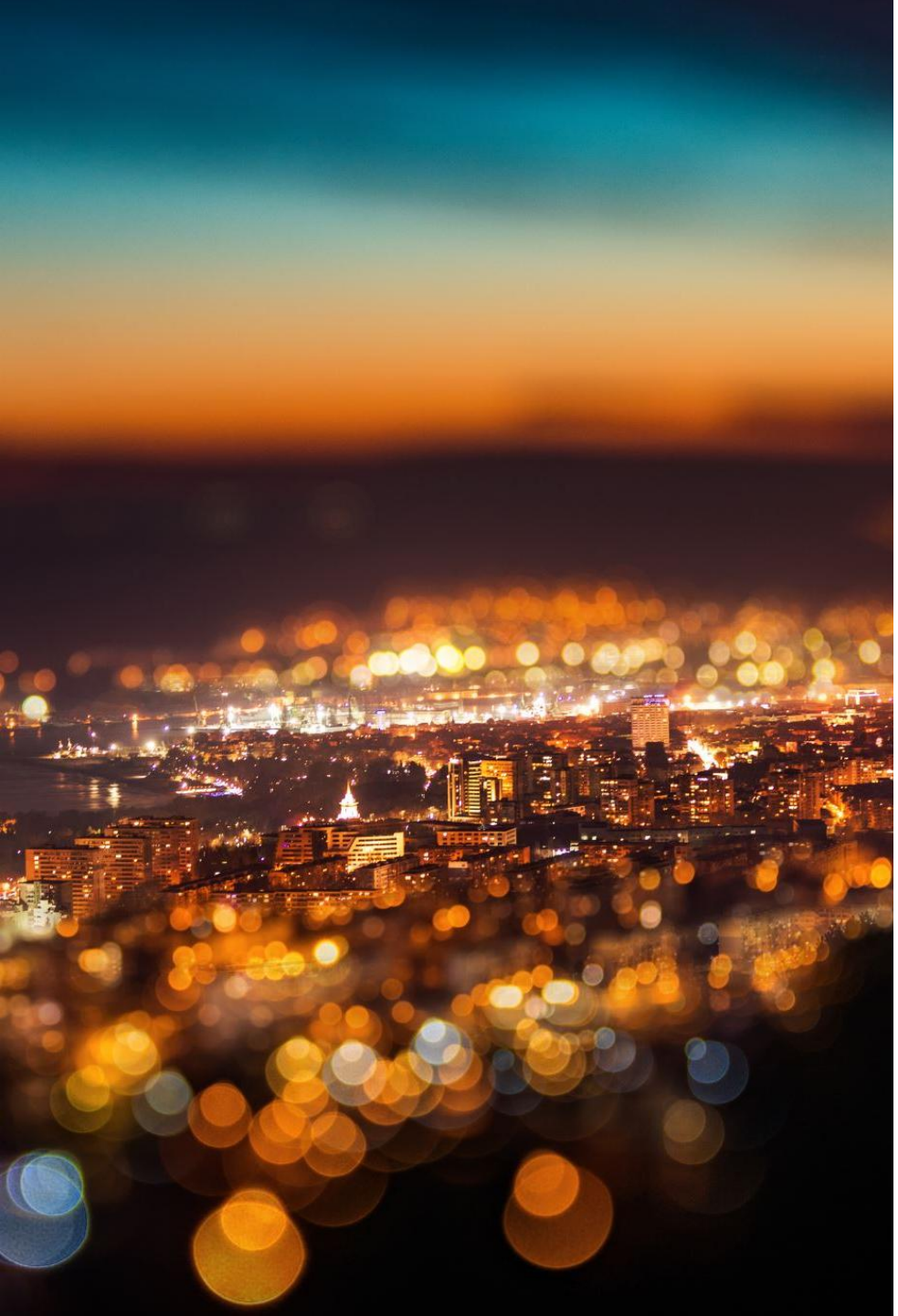


Demography

PROF. DR. FOUAD METRY ATIA

PROFESSOR OF COMMUNITY MEDICINE,

Demography

Demography is the scientific study of human populations. It is mainly concerned with

- **Size : It refers to the total number of persons in the given population.**
- **Distribution : It refers to the arrangement of entire population with respect to the geographical areas at a given point of time.**

Structure : It refers to the distribution of the given population with respect to age and sex.

Change : It refers to the increase or decrease in the size of the given population due to fertility, mortality and migration.

Development : It refers to the development of the given population with respect to socio economic aspects.

Other characteristic like genetic inheritance, intelligence and health.

Definition of Key Words

- **Size-Distribution-Structure-Change-Development**
- **Net-Migration refers to the total number of persons added or subtracted from a population as a result of the combined effect of immigration & emigration**

- **Urbanization** refers to mass migration of rural population into urban centers resulting in increasing the urban population & growth of cities
- **Population Growth Rate** (%/ year) is the rate at which a given population is expected to increase in a given period of time.

(1.5-1.9% in Egypt)

- **Population Doubling Time** refers to the time that would take for a population to double.
  - Because the population increase is on the compound interest formula, a 1% per year increase would cause the population to double in about 70 years.
  - If the population is increasing at 3% per year, than the doubling time will be 23.3 years
- **Population Pyramid** is a pictorial presentation of the age-sex composition of a population

Dependency Ratio

$$= \frac{\text{Children} + \text{Elderly}}{\text{Working Age}} \times 100$$

$$= \frac{\text{Pop below 15} + \text{65 \& above}}{\text{Pop 15} - \text{64 years}} \times 100$$

In 2026, the age dependency ratio in Egypt was 57.5% percent, which means that roughly 58 people were among the age groups 0 to 14 years and 65 years and older per 100 working-age population

Sex Ratio:

The overall sex ratio is simply the ratio of males to females in the population & is calculated by taking the number of males in a population & dividing it by the number of females in the same population

$$\text{Sex Ratio} = \frac{\text{Number of Males}}{\text{Number of Females}} \times 100$$

Sources of Demographic Data

The following are the sources of demographic data.

1. **Census**
2. **Surveys**
3. **Vital Events Registers**
4. **Sample Registration System**

Census

Census is a process concerned with:

- **Enumeration of individuals all over the country, at a certain time.**
- **Collection of demographic data and socioeconomic data of population; e.g.; age, sex, income, occupation, religious, education, ……. Etc.**

Types of census:

There are two methods of census taking

- **De facto: counting all persons in the area where they are present on the date of census irrespective of the person's nationality or his usual residence.**
- **De jure: when count is made, the population of each area being defined as persons who usually reside in the area, regardless of their actual location on the census date.**

- The de facto concept is much easier to apply and it is used in Egypt and England while in U.S.A the de jure basis is applied.
- The census is done every 10 years in Egypt

Egypt Median Age

Total	Male	Female
23.9	23.6	24.2

Egypt Population by Age

There are 63,632,655 people over age 18 in Egypt.

Census Years

Year	Date
1986	18 November 1986
1996	19 November 1996
2006	11 December 2006
2016	November 2016

Errors occurring during census:

Errors in numbers:

- Result from duplication of registration of a person in the census or not being registered at all.
- This results from changing the place at time of census.

Errors in data:

These result from:

- **Untrained data collector.**
- **Lack of awareness of the population about importance and accuracy of collected data.**

Uses of Census:

- Gives characteristic features of the population.
- Provides the needed data to calculate the statistical rates, and planning of different community programs (education, health services, and others)

Census and estimated population:

- **Census population:**
  - **number of population in the year of census taking.**
- **Estimated population:**
  - **number of population in any of the inter-census years.**

Methods of estimation:

1- Natural increase method:

The differences between number of live births and deaths in the year, which follow the census year, are added to the census population to get estimated population of a given inter-census year.

2. Arithmetic Growth Method

In this method it is assumed that there is an equal addition every year to the population during the intercensal period and this addition is taken to be average increase per year.

Two consecutive census populations of a given area are taken. e.g. 20 million in year 1990 and 40 million in year 2000.

Final estimated population in year 2004 may be estimated as follows:

Mean annual increase = 40 million –20 million / 10 = 2 million

Estimated population of 2004 = 40 million + (2 million x 4) = 48 million.

3. Graphic method:

A number of successive census populations are plotted on a graph and joined together by straight line, which is then extended over future years. Obtained graph can be used to find out:

- **Inter-census population of a given year within the censuses.**
- **Expected population of future years, on assumption that growth rate of the population will not change.**

4. Geometric Growth Method :

It is the most accurate method of estimation.

Special formula is used to find out the annual rate population of growth, to be applied to last census population, to get the estimated population of a given year.

5. Exponential Growth Method :

In this method it is assumed that there is an exponential growth.

6. Component Projection Method

This method is mainly used for future population projections (future estimates) using special model.

Demographic transitions

In 1929 the American demographer Warren Thompson, observed changes in birth and death rates in industrialized societies over the past two hundred years or so and then formulated a model called "Demographic Transition" that describes population change over time in fully developed country today, such as The United States or Canada, the countries of Europe, or similar societies elsewhere (e.g. Japan, Australia etc.).

Demographic Transition model recognizes five demographic stages namely

- high stationary,
- early expanding,
- late expanding,
- low stationary and
- declining.

It may be mentioned that sometimes these five stages are described as components of a so called "demographic cycle."

1. First stage (High stationary stage): This is the stage of high birth and death rates, which nearly balance each other. The population remains rather stationary.

2. Second stage (Early expanding): Death rate declines while fertility remains unaltered. The result is rapid population increase.

3. **Third stage (Late expanding):** Birth rate starts declining, while death rate falls still further. Population continues to increase.

4. **Fourth stage (Low stationary):** This is the stage of stable population when birth and death rates are low and balance each other.

5. **Fifth stage (Declining):** Birth rate is lower than the death rate. Population declines.

(Egypt in Third (Late expanding): stage)

Population Pyramid

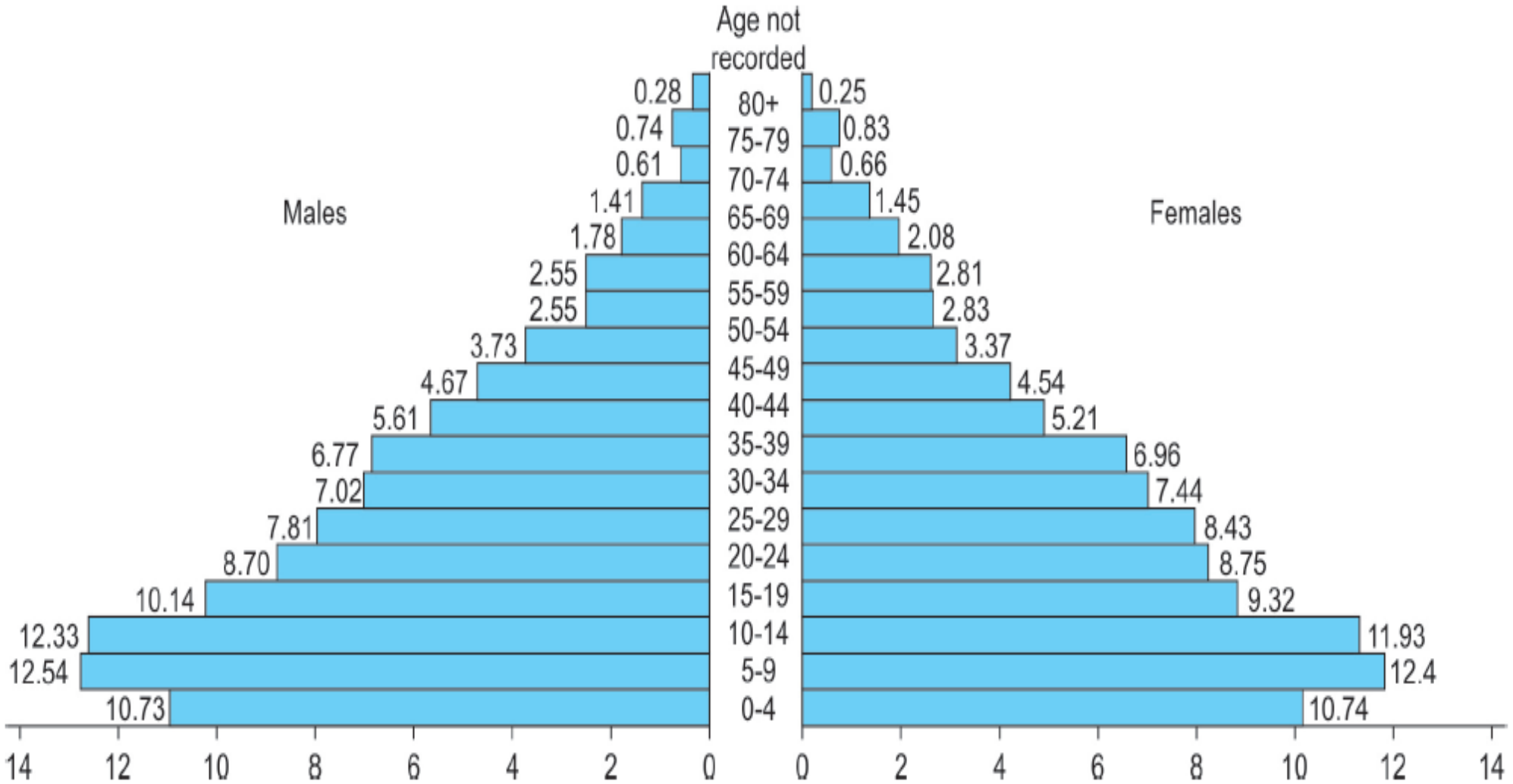
When the breakdown of the population is represented either as numbers or age percentages, according to different ages and sexes, in the form of

- **a horizontal bar diagram, horizontal axis referring to sex (men on the right side and women on left side) and**
- **vertical axis to age,**

it takes the shape of a 'Pyramid', which is called 'Population pyramid' (or 'Age-sex pyramid')

It shows the following data:

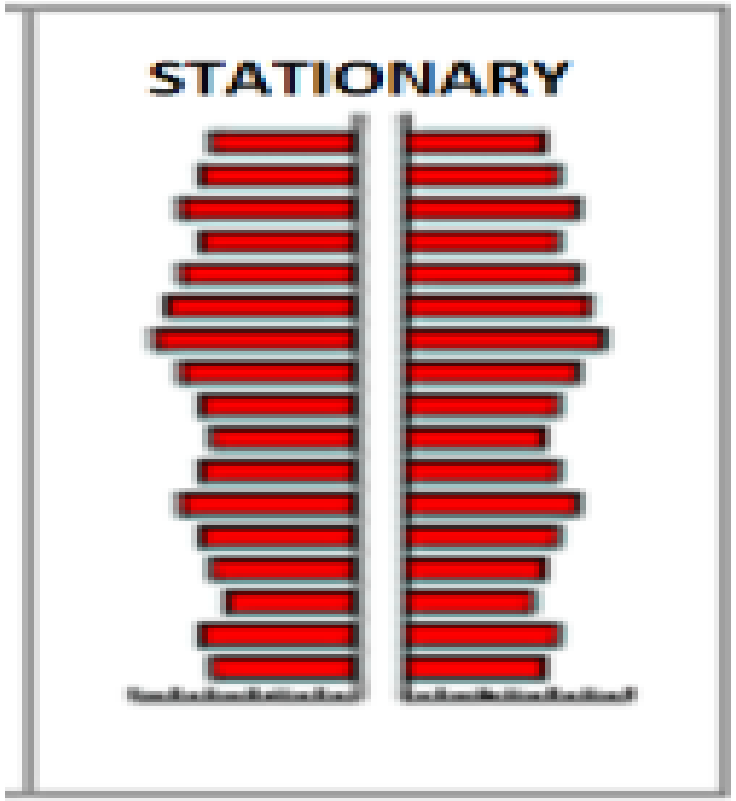
- Percentage of each age group (young and old)
- Sex ratio (percentage of old females)
- Character or pattern: Wide base and rapid tapering edges, narrow or pointed apex in developing countries, and nearly parallel edges and flat apex in developed countries


Age not recorded
Males
Females
80+
75-79
70-74
65-69
60-64
55-59
50-54
45-49
40-44
35-39
30-34
25-29
20-24
15-19
10-14
5-9
0-4
0.28
0.74
0.61
1.41
1.78
2.55
2.55
3.73
4.67
5.61
6.77
7.02
7.81
8.70
10.14
12.33
12.54
10.73
0.25
0.83
0.66
1.45
2.08
2.81
2.83
3.37
4.54
5.21
6.96
7.44
8.43
8.75
9.32
11.93
12.4
10.74
14 12 10 8 6 4 2 0
0 2 4 6 8 10 12 14

Types of population pyramid

"**Stationary" pyramid or constant population pyramid:**

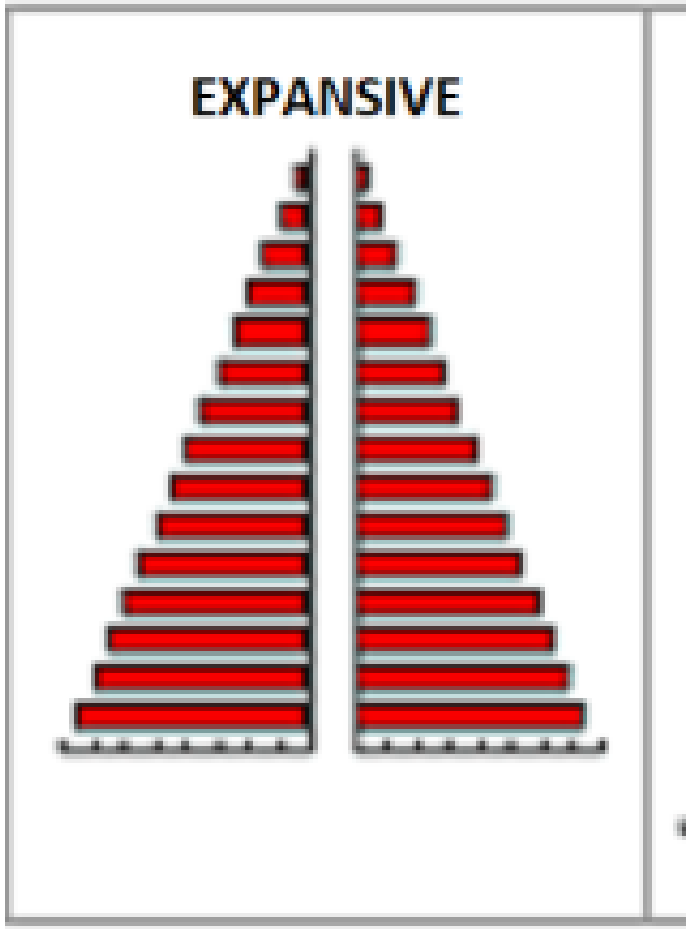
- A pyramid can be described as stationary if the percentages of population (age and sex) remain approximately constant over time.
- In a stationary population, the numbers of births and death roughly balance one another.



"Expansive" pyramid or Expanding population pyramid:

A population pyramid that is very wide at the younger ages, characteristic of countries with a high birth rate and perhaps low life expectancy.

The population is said to be fast-growing, and the size of each birth cohort increases each year.

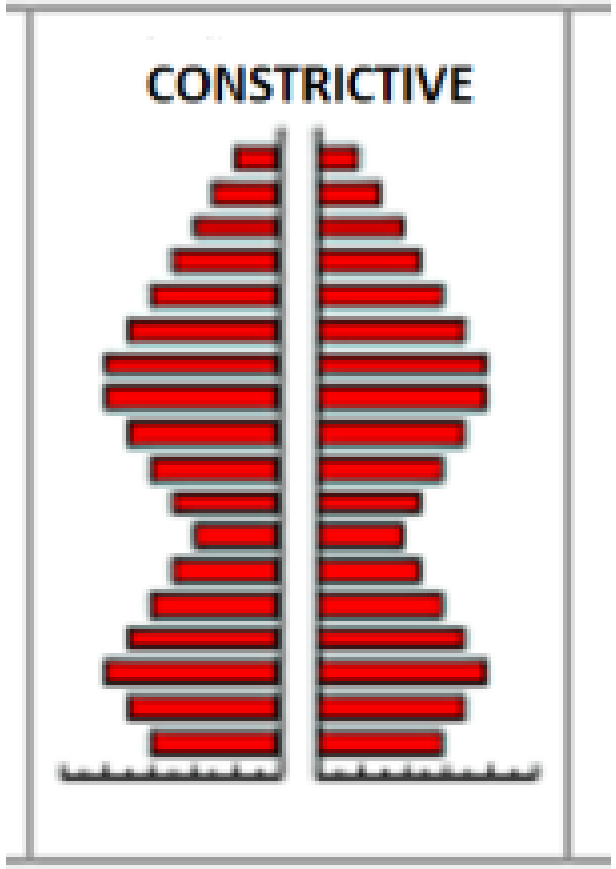


"Constrictive" pyramid or Declining population:

A population pyramid that is narrowed at the bottom. The population is generally older on average, as the country has long life expectancy, a low death rate, but also a low birth rate.

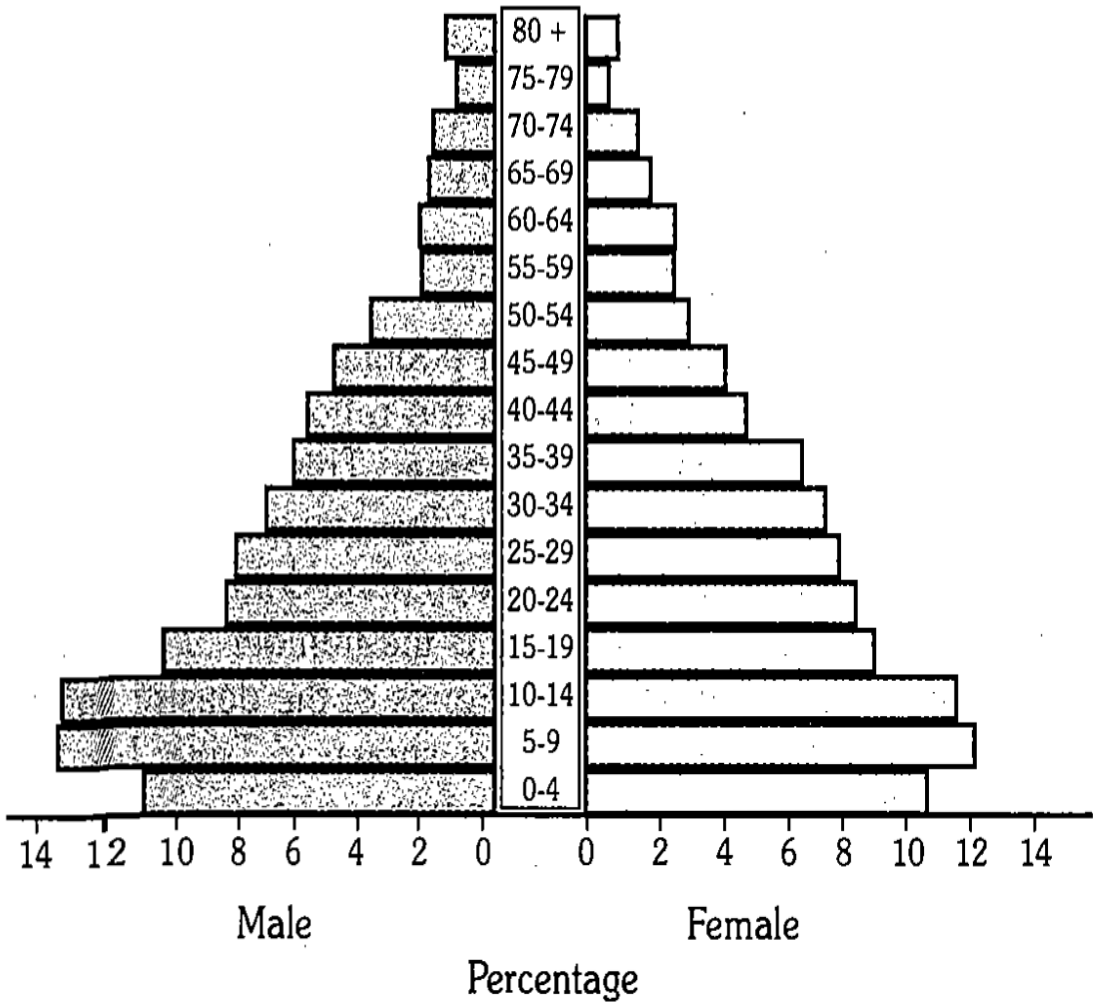
This may suggest that in future there may be a high dependency ratio due to reducing numbers at working ages.

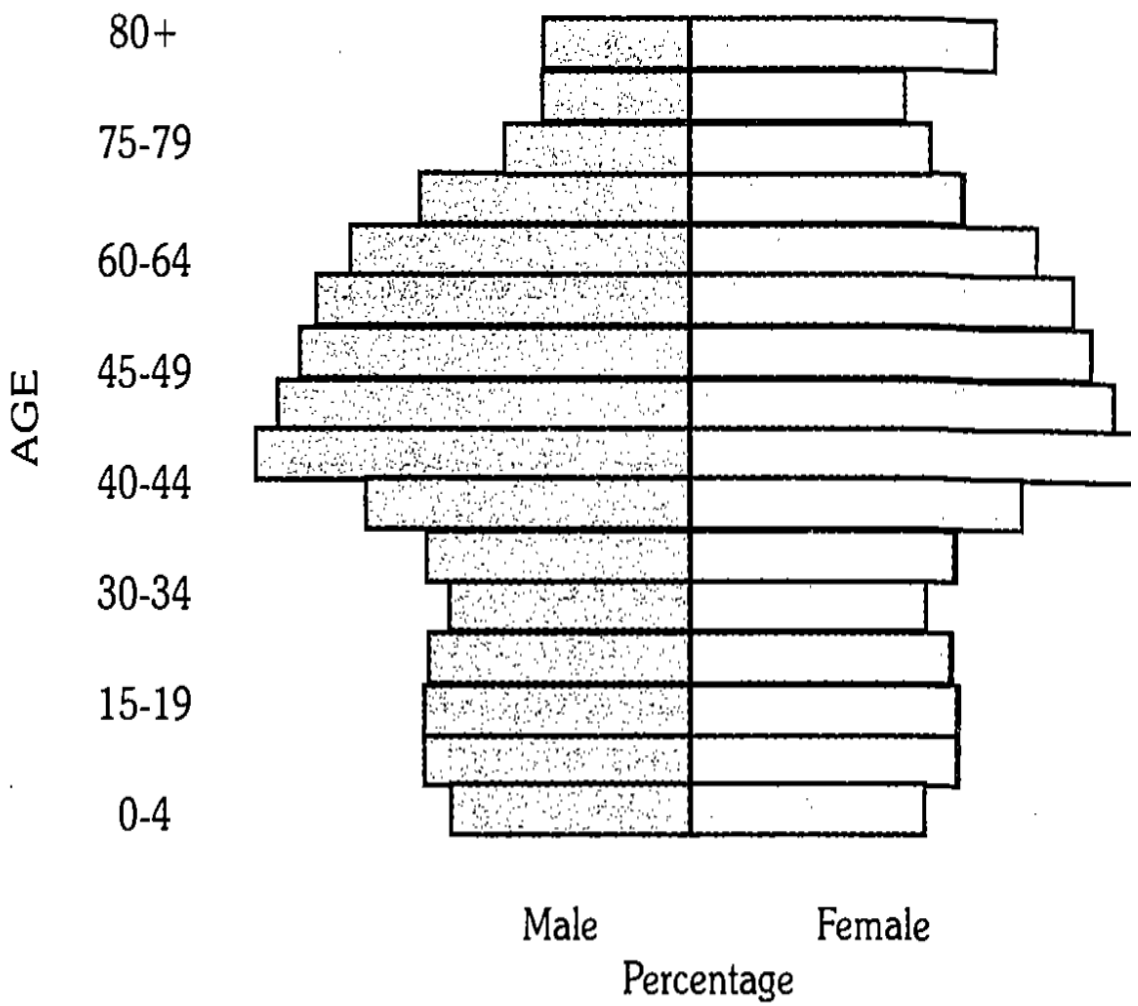
This is a typical pattern for a very developed country, with a high level of education, easy access to and incentive to use birth control, good health care, and few negative environmental factors.



Differences between the population pyramids of developing and developed countries

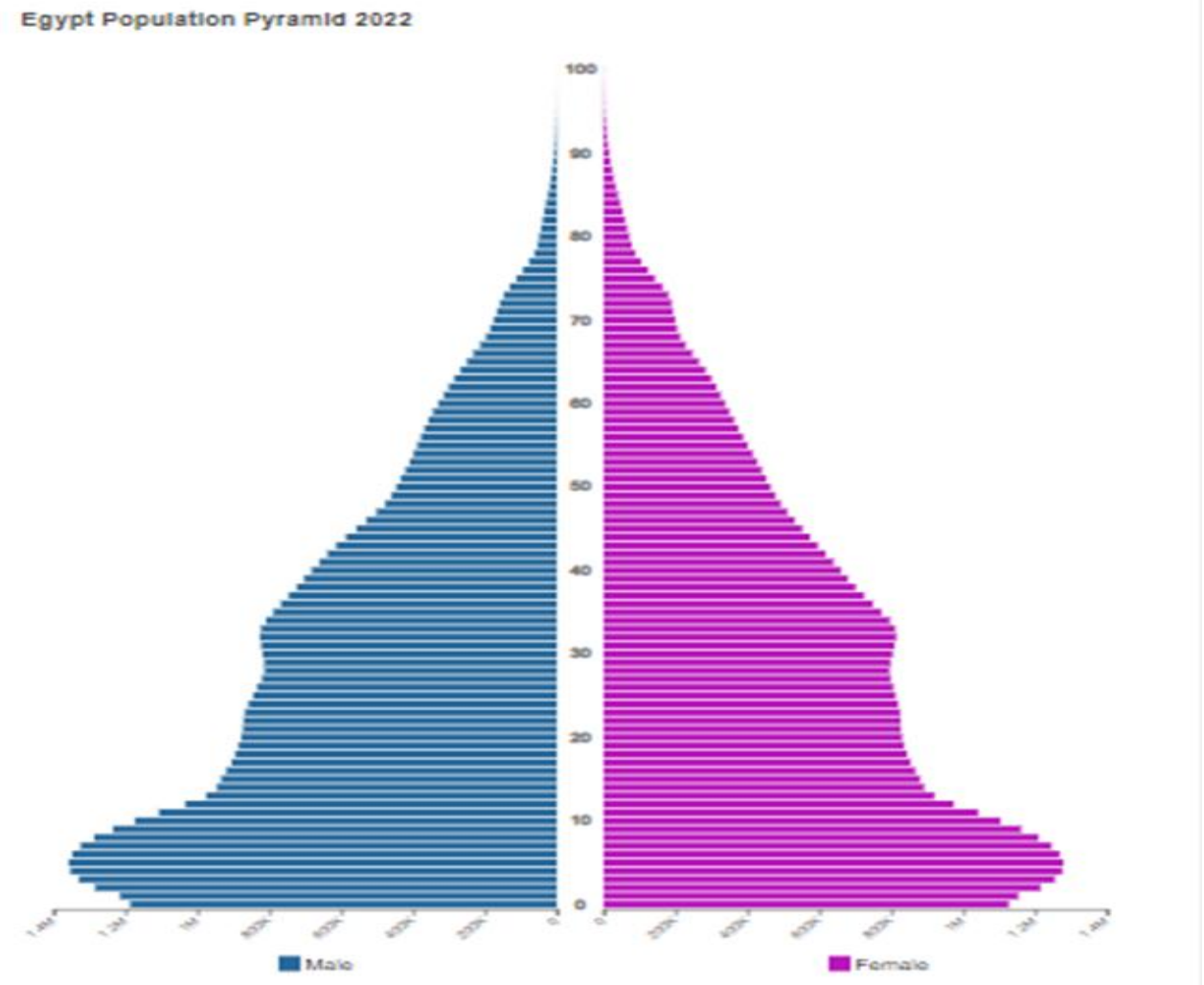
Developing country	Developed country
Broad base (indicating high birth rate)	Narrow base (low birth rate)
Concave border (concavity facing outwards)	Convex border (facing outwards)
Acute apex (indicating less number of elderly people)	Obtuse apex (more of elderly people)


INDIA (2001)
80 +
75-79
70-74
65-69
60-64
55-59
50-54
45-49
40-44
35-39
30-34
25-29
20-24
15-19
10-14
5-9
0-4
14 12 10 8 6 4 2 0
0 2 4 6 8 10 12 14
Male
Female
Percentage

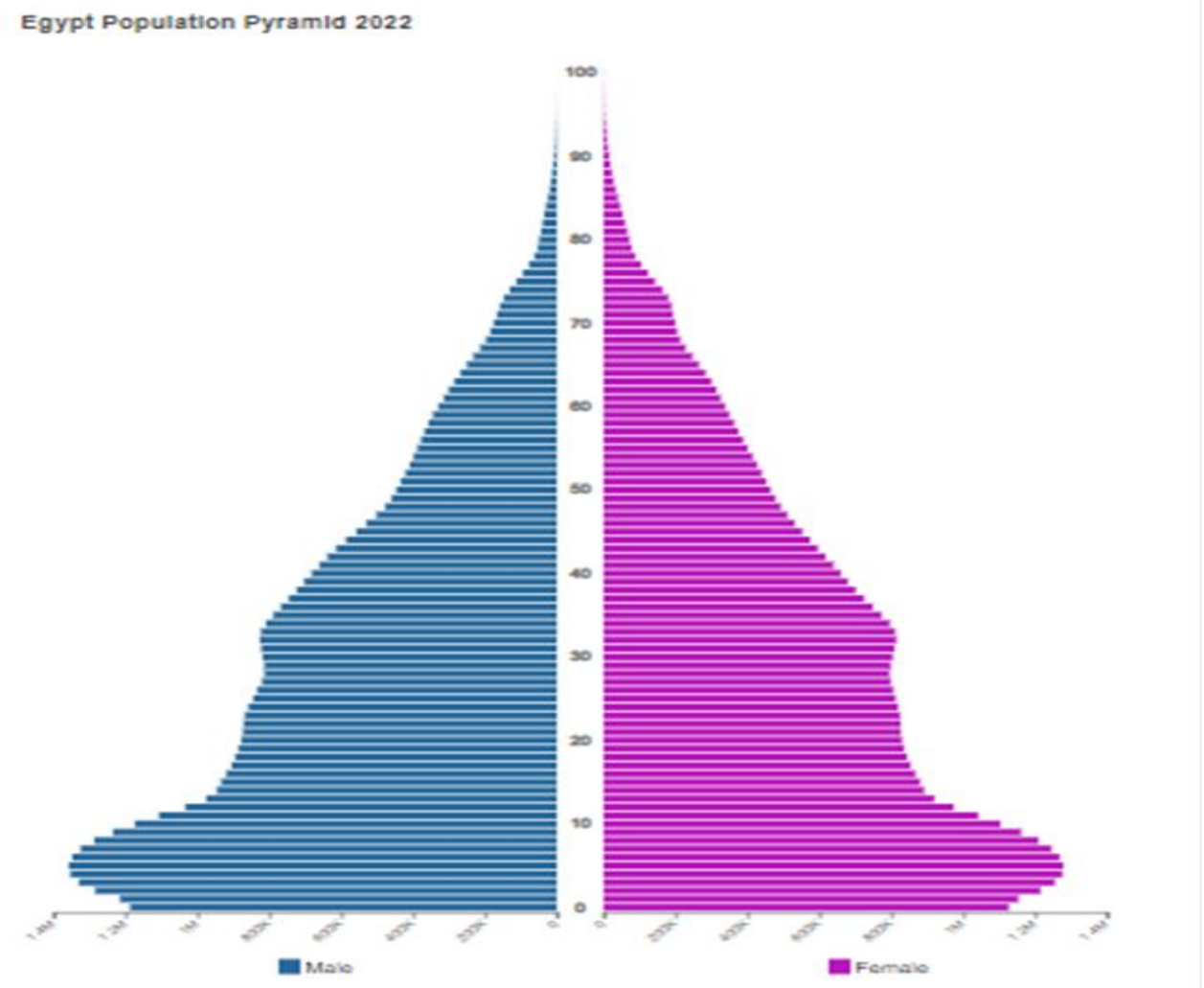

SWITZERLAND (2001)
AGE
80+
75-79
60-64
45-49
40-44
30-34
15-19
0-4
Male
Female
Percentage

Characteristics of population pyramid of Egypt:

- **Base layer,** representing age groups below 15 years, are broad due to high birth rate.
- **Top** of the pyramid is narrow, due to smaller proportion of the elderly aged 60 and over years.



- **Body** of the pyramid: are shorter than those of developed countries due to less mean life expectancy.
- **Sex distribution**: male/female ratio of the population shows insignificant difference with slightly more percent of males (51/49).



Thank you